11.2 Biofertilizers

In India, the availability and affordability of fossil fuel based chemical fertilizers at the farm level have been ensured only through imports and subsidies. Indiscriminate and imbalanced use of chemical fertilizers, especially urea, along with chemical pesticides and unavailability of organic manures has led to considerable reduction in soil health. Biofertilizers can act as a renewable supplement to chemical fertilizers and organic manures. They have the capacity to produce natural resistance in plants against pests and soil borne diseases, adding fertility to soil.

Nitrogen fixation by leguminous and other crops is reported to be 44 million metric tons per annum. The appropriate strain of *Rhizobium* can increase the crop yield up to 10-35%. Also, residual N is beneficial for the next crops grown in the same field.

It has been estimated that 40-250 kg N / hectare(ha) / year is fixed by different legume crops by the microbial activities of *Rhizobium*.

Definition

Biofertilizers are preparations containing beneficial micro organisms like N_2 fixers, PO_4 solubilizers in a viable static state intended for seed or soil application and designed to improve soil fertility.

Advantages

1. They reduce the need for chemical fertilizers.
2. They provide the plant with certain vitamins, plant growth promoting substances and increase the vigour of the plant.
3. It is cheap and cost effective.

Based on the nutrients that they provide, biofertilizers are of the following types

Nitrogenous biofertilizers-

- *Rhizobium,*
- *Azotobacter,*
- *Azospirillum*
- *Frankia*

Phosphate solubilisers

- *Bacillus*
- VAM

11.2.1 Rhizobium

Rhizobium – legume symbiosis is a well studied plant microbe interaction and *Rhizobium* is the most extensively used nitrogenous biofertilizer in India.

Rhizobium is a gram negative, non-spore forming aerobic bacillus inhabiting the soil in a free living state. The colonies of *Rhizobium* on YEMA (Yeast Extract Mannitol Agar) plate are gummy, pale white in colour (Figure 11.7). They can establish symbiotic relationship with leguminous plants and fix atmospheric nitrogen thereby greatly improving soil fertility.

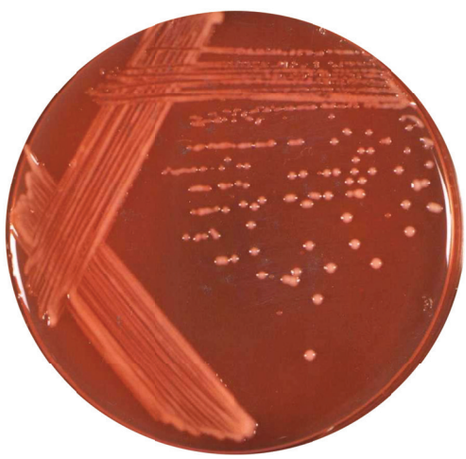


Figure 11.7: Pale pink mucoid colonies of *Rhizobium* on Yeast Extract Mannitol Agar plate

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